

STATISTICAL INFERENCE : TESTING OF HYPOTHESES

16.1 INTRODUCTION

Testing of Hypotheses may be described as a procedure in which sample data are used to decide whether or not to reject a proposition made about a population characteristic (parameter). Before introducing the formal statistical procedure, some basic concepts are defined first.

16.2 STATISTICAL HYPOTHESIS

A statistical hypothesis is a statement about a population parameter whose validity is to be tested on the basis of a random sample drawn from the population. The statement may be an assumption, a proposition, a claim, a preconceived idea, an educated guess, a conjecture or an assertion about any population parameter.

16.3 NULL AND ALTERNATIVE HYPOTHESES

A statistical hypothesis setup for possible rejection is called a Null Hypothesis and is denoted by H_0 . A hypothesis opposite to the null hypothesis is called an alternative hypothesis and is denoted by H_A . The objective to set a null and an alternative hypothesis is to gain support for alternative hypothesis by showing that the data disagree with the null hypothesis.

What is desired to be concluded about the population parameter is stated in the alternative hypothesis because it is accepted if the sample result does not support the null hypothesis.

The null hypothesis is always expressed as an equality and it is assumed to be true throughout the testing process.

There are three types of alternative hypothesis.

- (i) Population parameter is less than some specified value.
- (ii) Population parameter is more than some specified value.
- (iii) Population parameter is not equal to some specified value.

The first two are called one sided alternatives and the third one is a two sided alternative.

16.4 ONE AND TWO TAILED TESTS

A test procedure is called a one tailed test procedure if the alternative hypothesis is one sided. The test will be two tailed if the alternative hypothesis is two sided.

The following example illustrates the construction of null and the appropriate alternative hypothesis.

Example 16.3.1

Let a specified value of population mean is 45. Construct the null and alternative hypothesis for the following questions.

- (a) Do the sample data provide sufficient evidence to indicate that the population mean is greater than 45.
- (b) Do the sample data provide sufficient evidence to indicate that the population mean is less than 45.
- (c) Do the sample data provide sufficient evidence to indicate that the population mean is not equal to 45.

Solution :

The specified value of μ is called the hypothesized value of μ and is denoted by $\mu_0 = 45$. Thus,

- | | |
|---------------------|--------------------------|
| a) $H_0 : \mu = 45$ | leads to one tailed test |
| $H_A : \mu > 45$ | (or right tailed test) |
| b) $H_0 : \mu = 45$ | leads to one tailed test |
| $H_A : \mu < 45$ | (or left tailed test) |
| c) $H_0 : \mu = 45$ | leads to one tailed test |
| $H_A : \mu \neq 45$ | (or both tailed test) |

In each of these cases the null hypothesis has been expressed as $\mu = 45$ even the opposite of $\mu > 45$ is $\mu \leq 45$ and the opposite of $\mu < 45$ is $\mu \geq 45$. This is because a single value of μ is needed for the sampling distribution of $\bar{x}$ which is used to determine those value of $\bar{x}$ that contradict the null hypothesis and support the alternative hypothesis.

16.5 TYPE I AND TYPE II ERROR

Hypothesis testing is an uncertain process and may not always lead to correct decisions because sample data are used to reach conclusions about a population parameter. There are two types of incorrect decisions that can be made. One is the rejection of a true null hypothesis and the other is the acceptance of a false null hypothesis. The incorrect decision of rejecting a null hypothesis when it is true is called a type I error or error of first kind whereas the incorrect decision of accepting a null hypothesis when it is false is called a type II error or error of second kind.

16.6 LEVEL OF SIGNIFICANCE AND POWER OF A TEST

To reflect the uncertainty associated with the hypothesis testing process, the two types of errors are quantified by associating probabilities with them.

The probability of making type I error is called the level of significance of the test denoted by α .

The probability of making a type two error is denoted by β and $(1 - \beta)$ is called the power of the test.

Thus the power of a test is the probability of rejecting a false null hypothesis which is a correct decision. The table given below outlines the correct and incorrect decisions in hypotheses testing.

decision of the test for the Null Hypothesis	The Null Hypothesis is	
	True	False
Accept	Correct decision $P(\text{Correct decision}) = 1 - \alpha$	Incorrect decision type II error $P(\text{Type II error}) = \beta$
Reject	Incorrect decision $P(\text{Type I error}) = \alpha$ $= \text{level of significance of the test.}$	Correct decision $P(\text{correct decision}) = 1 - \beta = \text{Power of the test}$

In a hypothesis testing problem it is desirable to minimize both types of errors simultaneously, but decreasing one type of error may very likely increase the other type of error for a fixed sample size. However both types of errors can be reduced by increasing the sample size.

In practice the probability of type I error is kept fixed at a specified value and then type II error is minimized. A value of α such as 0.01, 0.02, 0.05, or 0.10 is usually fixed before taking a sample.

The probability of Type II error, β , depends on the true value of population parameter. Since it can have many possible values, it is necessary to examine the behaviour of β for several values of the parameter.

Example 16.6.1

A random sample of size 25 is drawn from a normal population with known $\sigma = 10$. Let α is fixed as 0.05.

For $H_0 : \mu = 30$

$H_A : \mu \neq 30$

Compute the values of β for

$$\mu = 22, 24, 26, 28, 30, 32, 34, 36, \& 38$$

Solution :

$$\begin{aligned}\text{Given : } \alpha &= P(\text{Type I error}) \\ &= P(\text{Reject } H_0 / H_0 \text{ is true}) \\ &= P(\text{Reject } \mu = 30 / \mu = 30) \\ &= P(\text{Accept } \mu \neq 30 / \mu = 30) \\ &= 0.05\end{aligned}$$

$$\begin{aligned}\text{Required : } \beta &= P(\text{Type II error}) \\ &= P(\text{Accept } H_0 / H_0 \text{ is false}) \\ &= P(\text{Accept } \mu \neq 30 / \mu \neq 30)\end{aligned}$$

(i) Let $H_A : \mu = 22$

The sampling distribution $\bar{x}$ is normal with

$$\mu_{\bar{x}} = 30 \text{ (taking } H_0 : \mu = 30 \text{ as True)}$$

$$\text{and } \sigma_{\bar{x}} = \frac{10}{\sqrt{25}} = 2$$

$$\alpha = 0.05 \Rightarrow z = 1.96$$

The two limits $\bar{x}_1$ and $\bar{x}_2$ containing 95% area under this normal curve are thus computed as :

$$\bar{x}_1 = \mu_{\bar{x}} - z_{\alpha/2} \cdot \sigma_{\bar{x}} = 30 - 1.96 \times 2 = 26.08$$

$$\bar{x}_2 = \mu_{\bar{x}} + z_{\alpha/2} \cdot \sigma_{\bar{x}} = 30 + 1.96 \times 2 = 33.92$$

H_0 will be rejected if $\bar{x}$ is less than 26.08 or over 33.92

Now consider the sampling distribution of $\bar{x}$ when

$H_0 : \mu = 30$ is false and $H_A : \mu = 22$ is true)

The distribution will be normal with

$$\begin{aligned}\mu_{\bar{x}} &= 22 \text{ (taking } H_A : \mu = 22 \text{ as true)} \\ \text{and } \sigma_{\bar{x}} &= 2\end{aligned}$$

$H_0 : \mu = 30$ will be accepted if $\bar{x}$ comes out to be over 26.08 This would be an incorrect decision (type II error) since $\mu = 22$ is taken as true and $\mu = 30$ as false. In other words when $\bar{x}$ is over 26.08 a false null hypothesis will be accepted leading to a type II error.

Thus

$$\begin{aligned}
 \beta &= P(\text{Type II error}) \\
 &= P(\text{Accept a false null hypothesis}) \\
 &= P(\text{Accept } \mu = 30 / \mu = 22 \text{ is true}) \\
 &= P(\bar{x} \geq 26.08 / \mu = 22 \text{ is true}) \\
 &= P\left(\frac{\bar{x} - 22}{2} \geq \frac{26.08 - 22}{2}\right) \\
 &= P(z \geq 2.04) \\
 &= 0.0207
 \end{aligned}$$

The decision limits for $\bar{x}$ are shown in figure 16.6.1

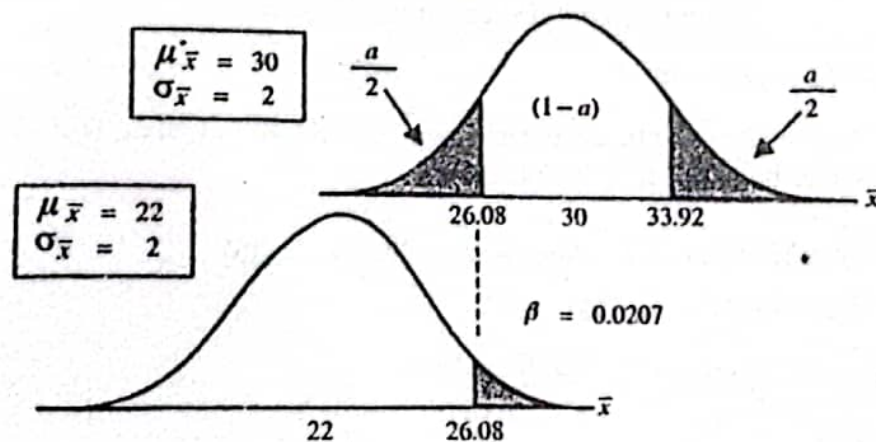


Figure 16.6.1 : the distribution of $\bar{x}$ under H_0 & H_A

($\mu = 22$) to determine β

(ii) Let $H_A : \mu = 28$

$H_0 : \mu = 30$ will be accepted when $26.08 \leq \bar{x} \leq 33.92$, but when $H_A : \mu = 28$ is true, the decision would be incorrect (Type II error).

$$\beta = P(26.08 \leq \bar{x} \leq 33.92 / \mu = 28)$$

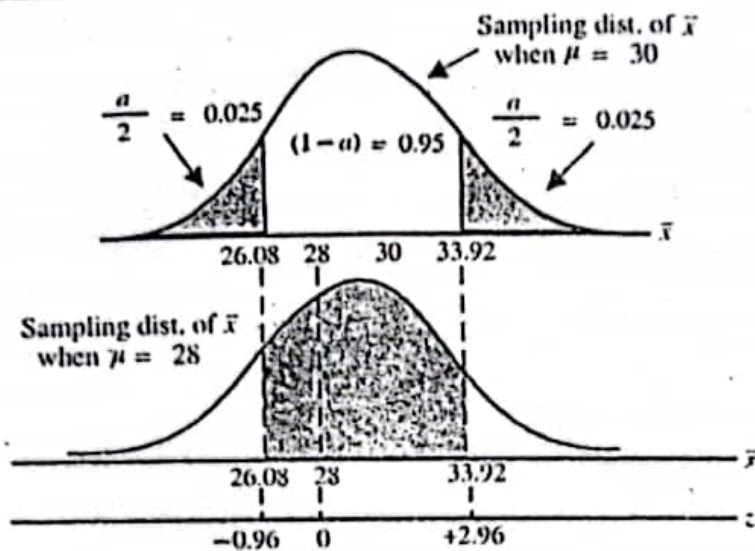
$$z_1 = \frac{26.08 - 28}{2} = -0.96$$

$$z_2 = \frac{33.92 - 28}{2} = +2.96$$

$$= P(-0.96 \leq z \leq +2.96)$$

$$= 0.81315 - 0.0015$$

$$= 0.8300$$

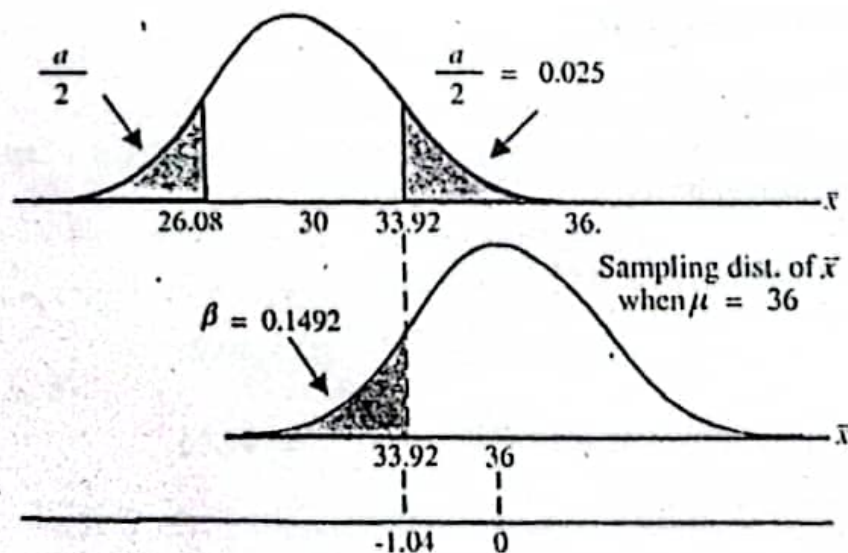
Figure 16.6.2 Determining β when $\mu = 28$

(iii) Let $H_A : \mu = 36$

$H_0 : \mu = 30$ will be accepted when $\bar{x} \leq 33.92$ but when $H_0 : \mu = 30$ is true the decision would be incorrect (Type II error).

$$\begin{aligned}
 \beta &= P(\text{Type II error}) = P(\text{Accept } \mu = 30 / \mu = 36) \\
 &= P(\bar{x} \leq 33.92 / \mu = 36) \\
 &= P(z \leq -1.04), z = \frac{33.92 - 36}{2} = -1.04 \\
 &= 0.1492
 \end{aligned}$$

Figure 16.6.3 shows this probability.

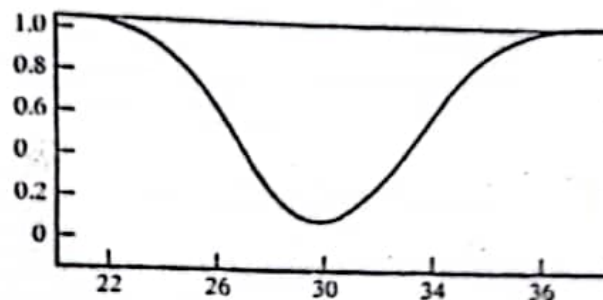
Figure 16.6.3 : determining β when $\mu = 36$.

In a similar manner the values of β can be determined for $\mu = 24, 26, 30, 32, 34$ and 38 . These values are tabulated below :

μ	β	$(\beta - 1)$
22	0.0207	0.9793
24	0.1492	0.8508
26	0.4840	0.5160
28	0.8300	0.1700
30	0.9500	0.0500
32	0.8300	0.1700
34	0.4840	0.5160
36	0.1492	0.8508
38	0.0207	0.9793

A curve drawn between μ and $(1 - \beta)$ is called a power curve. The power curve for the present example is shown in figure 16.6.4

Figure 16.6.4
Power curve for
Example 16.6.1



A curve drawn between μ and β is called an operating characteristic curve or an OC curve. An OC curve is the complement of the corresponding power curve.

16.7 TEST STATISTIC AND CRITICAL REGION

A test statistic is a quantity calculated from the sample that is used when making a decision about the hypothesis of interest. The numerical values of the test statistic for which the null hypothesis is rejected are called the critical values of the test and these values constitute a region called critical region or rejection region of the test.

The critical region is chosen so that the probability is α that it will contain the test statistic when the null hypothesis is true.

16.8 REJECTION RULE AND CONCLUSION

If the absolute value of the test statistic computed using sample data exceeds the absolute critical value of the test, the null hypothesis is rejected. A computed value of the test statistic higher than the critical value indicates that the difference between the sample estimate of the parameter and the hypothesized value of the parameter are significantly different resulting the rejection of the null hypothesis.

16.10 TESTING THE MEAN OF A POPULATION

1. Null Hypothesis : $H_0 : \mu = \mu_0$ (μ_0 is the hypothesized value of μ)

Alternative Hypothesis

- $H_A : \mu \neq \mu_0$ (Two tailed test)
 OR $H_A : \mu > \mu_0$ (One tailed test)
 OR $H_A : \mu < \mu_0$ (One tailed test)

2. Significance level :

$$\alpha = 0.01 \text{ OR } 0.05 \text{ OR } 0.10 \text{ etc.}$$

3. Test Statistic :

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \quad [\text{When } \sigma \text{ is known}]$$

$$\text{OR } z = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \quad [\text{When } \sigma \text{ is unknown and } n > 30]$$

$$\text{OR } t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \quad [\text{When } \sigma \text{ is unknown and } n \leq 30]$$

$$\text{with d.f.} = n - 1$$

The test statistic (z or t) is decided according to the following chart.

Sample Size (n)	Normal Population		Non-Normal Population	
	σ known	σ unknown	σ known	σ unknown
$n > 30$ large sample	z - test	z - test	z - test	None
$n \leq 30$ Small sample	z - test	t - test	None	None

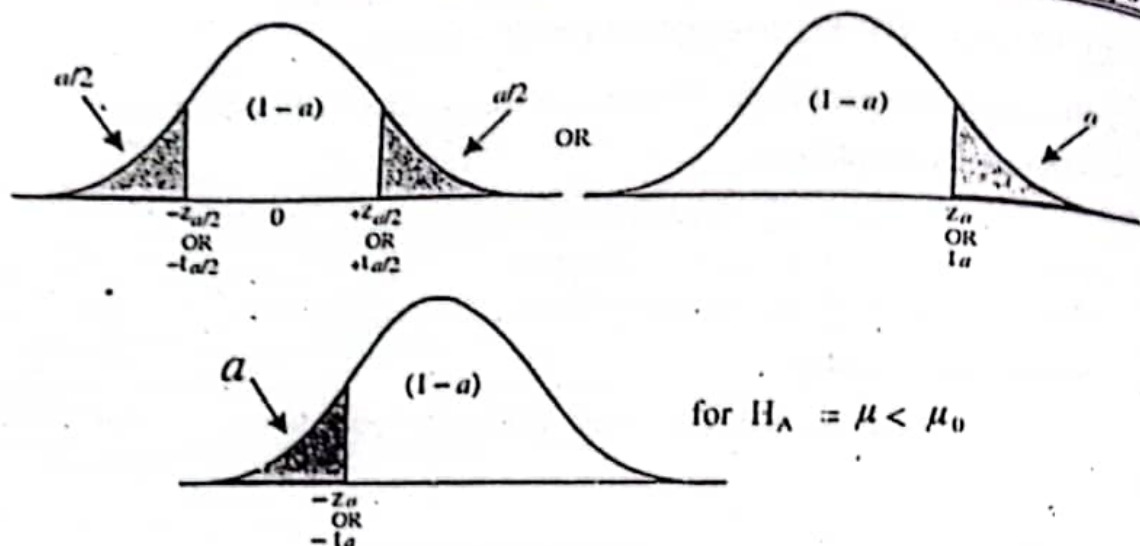
Thus t test is used only if

- (i) The population is normal AND
 (ii) σ is unknown (but s is known or can be computed) AND
 (iii) $n \leq 30$

4. Critical Region :

$$\text{for } H_A : \mu \neq \mu_0$$

$$\text{for } H_A : \mu > \mu_0$$



Critical values of z for commonly used values of α are:

Test	α	0.01	0.02	0.05	0.10
Two tail	$z_{\alpha/2}$	2.576	2.33	1.96	1.645
One tail	z_{α}	2.33	2.05	1.645	1.28

Critical values of t for $(n-1)$ degrees of freedom are obtained from the t -table. (Table C)

5. Rejection Rule :

If $|z_{cal}| > |z_{tab}|$, reject H_0 and accept H_A .
OR

If $|t_{cal}| > |t_{tab}|$, reject H_0 and accept H_A .

z_{cal} and t_{cal} are calculated values of the test statistic.

z_{tab} and t_{tab} are critical (or tabulated) values of the test statistic.

6. Conclusion :

Reject or do not reject the null hypothesis on the basis of the above rejection rule. The test will be significant if H_0 is rejected otherwise the test will be insignificant.

Example 16.10.1

A simple random sample of size 25 is taken from a normal population whose standard deviation is 4.

The sample mean is computed as 13.2

Do this result provides sufficient evidence to indicate that the population mean is more than 12 at 0.05 level of significance?

Solution :

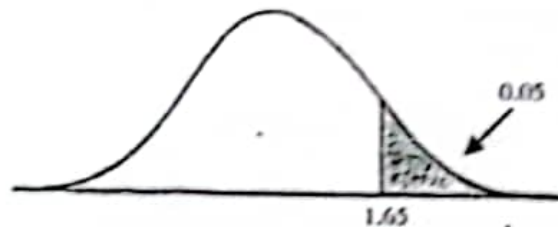
- Null hypothesis : $\mu = 12$
Alternative hypothesis : $\mu > 12$ (one tailed test)
- Significance level : $\alpha = 0.05$
- Test statistic : $z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$ (Since σ is known)

$n = 25$
$\bar{x} = 13.2$
$\mu_0 = 12$
$\sigma = 4$

$$z_{\text{cal}} = \frac{13.2 - 12}{\frac{4}{\sqrt{25}}} = 1.5$$

- Critical Region :

$$z_{\text{tab}} = z_{0.05} = 1.645$$



- Reject Rule :
Reject H_0 if $|z_{\text{cal}}| > |z_{\text{tab}}|$

- Conclusion :

Since $|z_{\text{cal}}| < |z_{\text{tab}}|$, H_0 can not be rejected, so H_A is rejected. The test is insignificant. It is thus concluded that the population mean μ is not more than 12.

Example 16.10.2

The mean operating temperature of a heat sensing device is claimed to be 190°F . A simple random sample of 16 devices has mean 195°F and standard deviation 8°F . Can it be concluded at the 0.01 level of significance that the mean operating temperature is higher than claimed?

Solution :

- Null hypothesis : $H_0: \mu = 190^\circ\text{F}$
Alternative hypothesis : $H_A: \mu > 190^\circ\text{F}$ (one tailed test)
- Level of Significance : $\alpha = 0.01$
- Test statistic :

$$t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} \quad (\text{Since } \sigma \text{ is unknown } < 30)$$

$$\text{d.f.} = 13$$

(s is used for unknown σ)

n	$= 15$
$\bar{X}$	$= 193$
μ_0	$= 190$
s	$= 8$

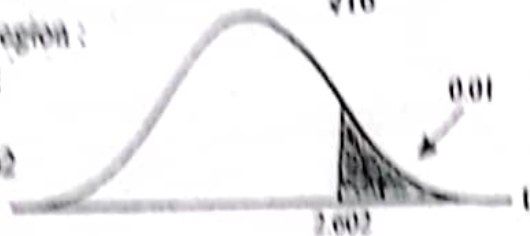
$$t_{\text{cal}} = \frac{193 - 190}{\frac{8}{\sqrt{15}}} = 2.5$$

4. Critical Region :

$$\alpha = 0.01$$

$$\text{d.f.} = 13$$

$$t_{\alpha/2} = 2.602$$



5. Rejection Rule :

$$\text{Reject } H_0 \text{ if } |t_{\text{cal}}| > |t_{\text{tab}}|$$

6. Conclusion :

$$|t_{\text{cal}}| = 2.5 \text{ and } |t_{\text{tab}}| = 2.602$$

Since $t_{\text{cal}} \not> t_{\text{tab}}$, H_0 cannot be rejected, so H_A is rejected. The test is insignificant. The mean temperature is not higher than claimed.

Example 16.10.3

A machine fills milk bottles, the mean amount of milk in each bottle is supposed to be 250 ml. In a routine check to see that the machine is operating properly, 36 filled bottles are chosen at random and found to contain a mean of 251.16 ml with a standard deviation of 2.4 ml.

At a 0.05 level of significance, is the machine operating properly?

Solution :

1. Null hypothesis : $H_0 : \mu = 250 \text{ ml}$

Alternative hypothesis : $H_A : \mu \neq 250 \text{ ml}$

(Two-tailed test)

2. Level of significance $\alpha = 0.05$

3. Test-statistic : $z = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} \quad (\text{Since } n > 30)$

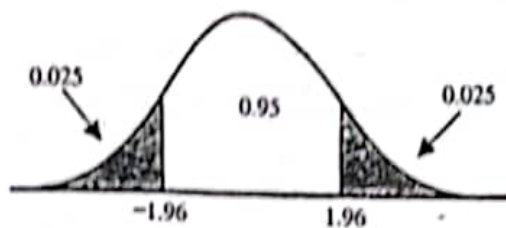
n	$= 36$
$\bar{X}$	$= 251.16$
μ_0	$= 250$
s	$= 2.4$

$$z_{\text{cal}} = \frac{251.16 - 250}{\frac{2.4}{\sqrt{36}}} = 2.9$$

4. Critical Region :

$$\alpha = 0.05$$

$$z_{\alpha/2} = 1.96$$



5. Rejection Rule :

Reject H_0 if $|\bar{x}| > |z_{\alpha/2}|$

6. Conclusion :

$$|z_{cal}| = 2.9 \text{ \& } |z_{tab}| = 1.96$$

Since $z_{cal} > z_{tab}$, reject H_0 and accept H_A .

The machine is not operating properly and it needs adjustments.

Example 16.10.4

An insurance company reports that the average annual maintenance cost for a Pakistan-made Suzuki car is currently Rs 3675. A random sample of 100 customers has mean annual maintenance cost of Rs. 3806 and standard deviation of Rs. 710. Is this result consistent with the company's report at the 0.05 level of significance ?

Solution :

1. Null hypothesis : $H_0 : \mu = \text{Rs. } 3675.$

Alternative hypothesis : $H_A : \mu \neq \text{Rs. } 3675$

2. Level of Significance : $\alpha = 0.05$

3. Test-statistics : $z = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \quad (\text{Since } n > 30)$

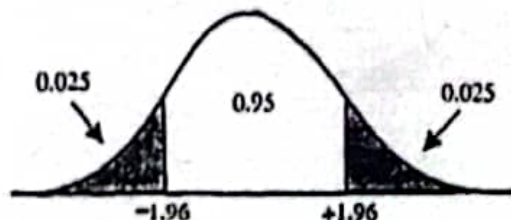
n	$= 100$
$\bar{x}$	$= 3806$
μ_0	$= 3675$
s	$= 710$

$$z_{cal} = \frac{3806 - 3675}{\frac{710}{\sqrt{100}}} = 1.845$$

4. Critical Region :

$$\alpha = 0.05$$

$$z_{\alpha/2} = 1.96$$



5. Rejection Rule :

Reject H_0 $\mu = 3675$ ml if $|z_{\text{cal}}| > |z_{\text{tab}}|$

6. Conclusion :

$$|z_{\text{cal}}| = 1.845 \text{ \& } |z_{\text{tab}}| = 1.96$$

Since $z_{\text{cal}} \neq z_{\text{tab}}$, H_0 can not be rejected.

H_A is then rejected. The sample result is therefore consistent with the company's report.

Example 16.10.5

The nicotine content (in milligrams) in a random sample of eight cigarettes of a certain brand are recorded as 3.9, 4.6, 4.5, 3.8, 2.1, 4.3, 5.1 and 5.3 mg.

Test the hypothesis that the average nicotine content does not exceed 3.5 mg as claimed by the manufacturer.

Assume the distribution of nicotine content as normal. Use 1% level of significance.

Solution :

1. Null hypothesis :

$$H_0 : \mu = 3.5$$

Alternative hypothesis :

$$H_A : \mu > 3.5$$

2. Level of Significance :

$$\alpha = 0.01$$

3. Test-statistic :

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

$$t_{\text{cal}} = \frac{4.2 - 3.5}{\frac{0.9957}{\sqrt{8}}}$$

$$\bar{x} = \frac{\sum x}{n} = \frac{33.6}{8} = 4.2$$

n	$= 8$
$\bar{x}$	$= 4.2$
μ_0	$= 3.5$
s	$= 0.9957$

$$s = \sqrt{\frac{n \sum x^2 - (\sum x)^2}{n(n-1)}}$$

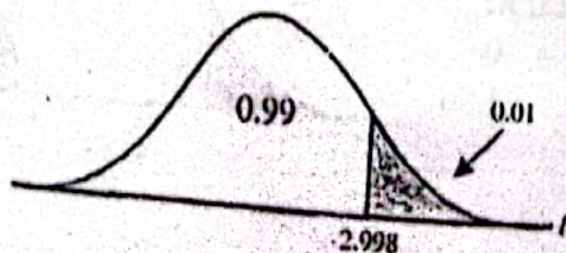
$$s = \sqrt{\frac{8(148.06) - (33.6)^2}{8 \times 7}} = 0.9957$$

4. Critical Region

$$\alpha = 0.01$$

$$df = 7$$

$$t_\alpha = 2.998$$



5. Rejection Rule :

If $|t_{cal}| > |t_{tab}|$, reject H_0

6. Conclusion :

$$|t_{cal}| = 1.99 \text{ \& \; } |t_{tab}| = 2.998$$

H_0 cannot be rejected as $t_{cal} < t_{tab}$

$\mu \leq 3.5$ may be accepted and thus it is concluded that the average nicotine content does not exceed 3.5 kg.

Example 16.10.6

A certain battery used in hand calculators is advertised as having a life of at least 500 hours. Student consumers test 25 of these batteries and find a mean life of 474 hours and standard deviation of 20 hours. Should the manufacturer's claim be accepted ? Use a 0.05 level of significance.

Solution :

1. Null hypothesis :

$$H_0 : \mu = 500 \text{ hrs.}$$

Alternative hypothesis :

$$H_A : \mu < 500 \text{ hrs. (one tailed test)}$$

2. Level of Significance :

$$\alpha = 0.05$$

3. Test-statistic :

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \quad [\sigma \text{ unknown and } n < 30]$$

$$n = 25$$

$$\bar{x} = 474 \text{ Hrs}$$

$$\mu_0 = 500 \text{ Hrs}$$

$$s = 20 \text{ Hrs.}$$

$$\text{d.f. } (n-1)$$

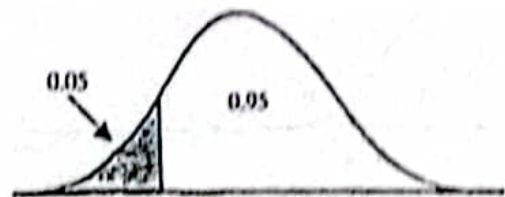
$$t_{cal} = \frac{474 - 500}{\frac{20}{\sqrt{25}}} = -6.5$$

4. Critical Region :

$$\alpha = 0.05$$

$$\text{d.f.} = 24$$

$$t_{\alpha} = -1.711$$



5. Rejection Rule.

If $|t_{cal}| > |t_{tab}|$, reject H_0

6. Conclusion :

$$|t_{cal}| = 6.5 \text{ \& \; } |t_{tab}| = 1.711$$

Since $|t_{cal}| > |t_{tab}|$, H_0 is rejected

Manufacturer's claim cannot be accepted and $\mu < 500$ Hrs.